Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks include reasoning, learning, decision-making, problem-solving, language understanding, and visual perception.

Machine Learning (ML) is a subset of Artificial Intelligence that enables systems to learn from data without being explicitly programmed. Instead of writing fixed rules, developers train machine learning models using datasets so the system can identify patterns and make predictions.

There are three main types of machine learning:

1. Supervised Learning

Supervised learning uses labeled data for training. In this approach, the model learns from input-output pairs. Examples include email spam detection, house price prediction, and image classification.

Common supervised learning algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- Support Vector Machines
- Random Forest

2. Unsupervised Learning

Unsupervised learning uses unlabeled data. The system identifies hidden patterns or structures in the data without predefined labels.

Examples:

- Customer segmentation
- Recommendation systems
- Market basket analysis

Common unsupervised learning algorithms:

- K-Means Clustering

- Hierarchical Clustering
- PCA (Principal Component Analysis)

3. Reinforcement Learning

Reinforcement learning is based on rewards and penalties. An agent interacts with an environment and learns the best actions to maximize rewards.

Applications include:

- Robotics
- Self-driving cars
- Game-playing AI

Deep Learning

Deep Learning is a specialized subset of machine learning based on artificial neural networks. It uses multiple layers of neurons to process complex patterns in data.

Deep learning performs exceptionally well in:

- Image recognition
- Speech recognition
- Natural language processing
- Autonomous systems

Popular deep learning frameworks:

- TensorFlow
- PyTorch
- Keras

Natural Language Processing (NLP)

Natural Language Processing enables computers to understand and process human language.

Applications of NLP:

- Chatbots

- Translation systems
- Sentiment analysis
- Voice assistants

Common NLP techniques:

- Tokenization
- Stemming
- Lemmatization
- Named Entity Recognition

Large Language Models (LLMs) like Llama, GPT, and Mistral are advanced NLP systems trained on massive datasets.

Vector Databases

Vector databases are designed to store embeddings or vector representations of data. They are essential for AI applications such as semantic search and Retrieval-Augmented Generation (RAG).

Popular vector databases:

- ChromaDB
- Pinecone
- FAISS
- Weaviate

ChromaDB is lightweight and suitable for local AI applications.

Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with language generation. Instead of relying only on the knowledge stored inside the language model, RAG retrieves relevant information from external documents before generating answers.

Steps in RAG:

1. Load documents
2. Split documents into chunks
3. Generate embeddings
4. Store embeddings in vector database
5. Retrieve relevant chunks
6. Send chunks to LLM
7. Generate final response

Benefits of RAG:

- Reduces hallucinations
- Improves accuracy
- Uses private data
- Supports document-based question answering

LangChain

LangChain is a framework for building applications powered by large language models. It simplifies the integration of LLMs with external data sources, APIs, and vector databases.

Main components of LangChain:

- Document loaders
- Text splitters
- Embeddings
- Retrievers
- Chains
- Memory

LangChain supports integration with Ollama, HuggingFace, ChromaDB, and many other AI tools.

Ollama

Ollama is a tool that allows users to run large language models locally on their own computer. It supports models such as:

- Llama 3
- Mistral
- Gemma

Benefits of Ollama:

- No API cost
- Privacy
- Offline usage
- Easy local deployment

Command to run Llama 3:

ollama run llama3

ChromaDB

ChromaDB is an open-source vector database commonly used in AI applications. It stores embeddings and enables fast semantic search.

Advantages:

- Lightweight
- Easy integration
- Local storage support
- Fast retrieval

ChromaDB is widely used in RAG applications.

HuggingFace Embeddings

HuggingFace provides pretrained transformer models for generating embeddings.

Popular embedding model:

- sentence-transformers/all-MiniLM-L6-v2

Embeddings convert text into numerical vector representations that capture semantic meaning.